

Athlete Kinetics: Concept & Architecture Verification

This document serves as a peer-review tool. It translates the software architecture back into biomechanical and engineering terms to verify that the system logic aligns exactly with your vision.

If any of the concepts below do not match your understanding of human kinematics or project goals, flag them, and the architecture will be adjusted.

1. The Core Engineering Philosophy

- **The Goal:** A closed-loop, local-only human performance tracker and prescriptive engine.
- **The Constraint:** Zero external APIs. Zero cloud compute costs. It must run entirely on the user's mobile device.
- **The Metaphor:** The app is a local operating system for the human body. It treats the user as a dynamic physics system subject to mechanical load (inputs) and physiological fatigue (outputs).

2. Physical-to-Digital Mapping (The Database)

We have mapped your physical training environment into three strict data domains:

- **Domain 1: Mechanical Input (Work Done)**
 - *Concept:* Tracking total force output and perceived exertion.
 - *Software Execution:* The app logs macro-cycles, sessions, sets, load (kg), and RPE. It strictly enforces the calculation of "Tonnage" at the database level so the math cannot be corrupted by UI glitches.
- **Domain 2: Telemetry (System Fatigue)**
 - *Concept:* Tracking the biological cost of the mechanical input.
 - *Software Execution:* A ring-buffer database that constantly ingests SpO2, HRV, and Sleep data from wearables.
- **Domain 3: The State Vector (The Kinematic Equation)**
 - *Concept:* Reducing all inputs and outputs to a single, readable readiness state.
 - *Software Execution:* Instead of the AI doing math on the fly, the database pre-calculates an Acute:Chronic Workload Ratio (ACWR) and a composite Readiness Score (combining HRV, load, sleep, and SpO2).

3. The Edge AI Logic (The "Coach")

Because we are running the AI locally on a phone, we cannot treat it like a conversational chatbot (e.g., ChatGPT). That would consume too much memory and lead to "hallucinated" or dangerous advice.

- **The Mechanism:** The AI acts as a strict, deterministic logic gate.
- **The Constraint (GBNF Grammar):** We have explicitly banned the AI from generating free-form text for training parameters. It is forced to select from a hard-coded menu of mechanical adjustments.
- **The Output:** It will only ever output four things:
 1. `load_modifier` (e.g., drop weight to 85%).

2. `set_modifier` (e.g., cut 2 working sets).
3. `rpe_cap` (e.g., do not exceed RPE 7.5).
4. `coaching_cue` (A single, blunt sentence explaining the mechanical rationale, like "HRV is severely suppressed; cap RPE to prevent central nervous system burnout").

4. The Hardware Reality (Memory Management)

You noted your interest in electrical engineering and hardware limits. Phones are hostile environments for AI due to "Jetsam" (the OS mechanism that kills apps using too much RAM).

- **Our Solution:** We have capped the AI at 1.5 Billion parameters for older phones (4GB RAM) and 3 Billion for newer ones (6GB+).
- **Resource Isolation:** The database is deliberately starved of cache memory (capped at 16MB, dropping to 4MB during AI inference) so it never fights the AI for physical RAM. When the app goes into the background, the AI context is immediately dumped from memory to prevent the OS from crashing the app.

Verification Checklist

Please review the following statements. If you agree with all of them, the software architecture is 100% aligned with your vision.

- ☐ I want the AI to be a strict mechanical calculator (adjusting sets/reps/load) rather than a conversational chat buddy.
- ☐ I agree with the composite readiness calculation prioritizing HRV, Load, Sleep, and SpO2.
- ☐ I accept that running this locally means the app will have strict memory management, instantly unloading the AI when I switch to another app to save battery and prevent crashes.
- ☐ I agree that the database should automatically calculate metrics like ACWR rather than relying on the AI to do the math.